MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) BCA EXAMINATION

Fourth SEMESTER

MG4CCRBCA200 - ARTIFICIAL INTELLIGENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer all questions
Each question carries 2 marks

1. Define an intelligent agent. [K] / [CO1]
2. Explain how a knowledge-based agent uses knowledge to make decisions? [U] / [CO1]
3. Describe how a genetic algorithm works. [U] / [CO2]
4. Explain propositional logic and its role in artificial intelligence. [U] / [CO3]
5. State the main goal of machine learning. [K] / [CO4]

[2x5 = 10]

Part B
Short Answer Questions
Answer any 5 questions
Each question carries 6 marks

6. Explain the structure of an intelligent agent with a neat diagram. [U] / CO1]
7. Define a problem-solving agent and state its role in artificial intelligence. [K] / CO1]
8. Explain the percepts available to an agent in the Wumpus World and their significance in decision-making. [U] / CO1]

9. Analyze the efficiency of alpha–beta pruning compared to the standard minimax algorithm. [An] / [CO2]

10. Explain the role of unification in the resolution process. [U] / CO3]

11. Illustrate how conditional independence reduces complexity in probabilistic reasoning. [U] / [CO3]

12. Outline the key differences between supervised and unsupervised learning with examples. [U] / [CO4]

[6x5 = 30]

Part C
Essay Questions
Answer any 2 questions
Each question carries 15 marks

13. Explain the AO* algorithm with an example. Discuss the handling of AND/OR nodes cost updating, and backtracking. [U] / [CO2]

14. Describe the forward chaining inference mechanism with a suitable example. [U] / [CO3]

15. Analyze the strengths and limitations of expert systems in real-world applications. [An] / [CO4]

[15x2 = 30]